

Massive Data Final Project 2
**Effect of Congestion Pricing Policy on For-Hire
Vehicles' Mobility in New York City**

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[**Github Repository**](#)

[**Project Website**](#)

Introduction & Background

Urban Congestion as a Policy Problem

Urban traffic congestion represents one of the most persistent, costly, and economic challenges facing large metropolitan areas, like New York City. In 2024, the average American lost 43 hours to traffic, the equivalent of an entire work week, costing the nation \$74 billion in lost time. (Sager, 2025) These costs are not distributed equally. Minority and low-income communities are disproportionately impacted by the noise and emissions generated by major roadways, additionally bearing a greater share of the environmental and public health burden created by dense vehicle traffic in the city.

New York City exemplifies this problem at scale. As one of the most congested urban areas in the United States, NYC drivers lost 101 hours to traffic jams in 2023, costing the city's economy \$9.1 billion. (Sager, 2025) In addition to the boom in ride-sharing apps, a 2021 study by the Singapore-MIT Alliance found that rideshare entry not only increases congestion but also discourages commuters from using greener alternatives, like walking, and showcased an 8.9 percent drop in public transport ridership. (Singapore-MIT Alliance for Research and Technology, 2021).

The environmental consequences of urban vehicle traffic are the primary focus of traffic intervention policies. The United States Environmental Protection Agency (EPA) reports that transportation accounts for over 28 percent of total U.S. greenhouse gas emissions. Within cities, the health impacts are measurable. Traffic-related air pollution has been causally linked to the onset of childhood asthma, lung, and cardiovascular diseases. (EPA, 2026) These harms are again unequally distributed: research funded by the EPA found that people of color breathe more particulate air pollution on average across income levels and regions, a pattern attributable to systemic factors that have historically co-located communities of color alongside high-emission corridors. In Manhattan, where vehicle density is extreme, and residential populations live close to major roadways, the intersection of congestion and environmental harm is extremely important and necessary to address.

Congestion Pricing as a Policy Tool

The theoretical rationale of congestion pricing is straightforward: a toll where the inner city congestion appears (in New York, Manhattan is the most densely congested) will reduce traffic. While this is the first of its kind to be implemented in the United States, other countries have utilized this tool and have seen striking results. Stockholm's congestion charge produced an immediate and lasting 20 percent reduction in traffic in its targeted zone, with overall congestion levels falling to 30 to 50 percent. (Kassirer & Boddy, 2014) London's experience was striking as well; traffic across London dropped an average of 30 percent, transit ridership increased, and overall air quality improved. Beyond traffic, congestion pricing in London, Stockholm, and Milan reduced CO₂, PM₁₀, and NO_x emissions by 13 to 18 percent within the congestion zones. (Crocì, 2016)

As a result, the Metropolitan Transportation Authority (MTA) and New York Governor Kathy Hochul launched the Congestion Relief Program, legislatively passed as the “Central Business District Tolling Program” (CBDTP) on January 5, 2025. The geographic boundaries of the Congestion Relief Zone were deliberately drawn to target Manhattan’s more economically dense and traffic corridors. This zone includes all regions past 60th Street: Midtown, the Financial District, SoHo, Tribeca, Chelsea, and all neighborhoods south of Central Park. (Metropolitan Transportation Authority, n.d.-a).

Private passenger vehicles are subject to a flat \$9 daily charge, while trucks and buses face higher rates ranging from \$14.40 to \$21.00 per entry. For vehicles licensed by the New York City Taxi and Limousine Commission, *Yellow taxis* are charged \$0.75 per trip and operate city-wide. *Green taxis* are also charged \$0.75 per trip, however, they are restricted to outer-borough and upper Manhattan, so their exposure to the CRZ is more limited by design. Lower volume *for-hire vehicles*, luxury limos, are charged \$0.75 per trip, but fall below 10,000 trips per day. The *high-volume for-hire vehicles*, exclusively Uber and Lyft, pay \$1.50 per trip, and operate at a large scale with over 10,000 trips per day.



Figure 1. Congestion Pricing Zone (MTA) Map

Research Contributions

This study examines the causal effect of congestion pricing on origin-destination trip volumes for taxis and for hire vehicles entering and leaving the CPZ. Specifically, it questions the causal effect of congestion pricing on taxis and for-hire vehicles' mobility patterns. While considerable research has examined the aggregate traffic effects of congestion pricing programs internationally, comparatively less work has focused primarily on taxi and rideshare activity.

For-hire vehicles are a particularly consequential subject of study under this policy for several reasons. Unlike private vehicles that pay a flat \$9 daily toll regardless of how many trips they make, taxis and for-hire vehicles are charged on a per-trip basis, with the cost passed directly to passengers. (MTA, n.d.) Furthermore, for-hire vehicles make repeated passes through the congested zone throughout the day, meaning their collective response to pricing has outsized implications for overall traffic volume in the CRZ. Using trip-level TLC data spanning January 2024 through December 2025, this study employs a difference-in-differences framework to estimate the causal effect of the CBDTP on origin-destination trip volumes across yellow taxis, green taxis, for hire vehicles (FHV), and high volume for hire vehicles (HVFHVs).

Data description

This study uses trip-level data from the New York City Taxi and Limousine Commission (TLC) portal for the following types of vehicles. Even though the data spans January 2009 - February 2026, the time period of the study is from January 2024 to December 2025. Specifically, the study used data related to two types of vehicles, which are:

- Yellow taxi (at least 3 million trips per month)
- High Volume For-Hire Vehicles (HVFHV) (at least 20 million trips per month)

In addition to yellow taxis and HVFHV, the TLC also has data for green taxis and other for-hire vehicles. However, these were excluded due to either data inconsistencies, or a low quantum of data. For each type of vehicle, the key variables can be divided into three main categories: temporal variables, spatial variables, and trip characteristics.

- **Temporal Variables:** This includes timestamps capturing the timing of trips including pickup datetime and dropoff datetime, which can be used to construct trip duration and time-of-day indicators.
- **Spatial Variables:** This includes trip origins and destinations identified using Taxi Zone IDs, such as pickup location and dropoff location. These locations map to 263 predefined taxi zones in New York City.
- **Trip Characteristics:** This includes measures of trip behavior and performance, such as trip distance, passenger count and fare-related components, including congestion charges.

Data processing begins by setting up a data pipeline that connects cloud storage (S3) to the data warehouse environment. This allows loading large parquet datasets and working with

them directly. From there, key variables are extracted and are cast into consistent data types, such as timestamps for trip times and numeric formats for distances and fares, so that the data is structured and usable for analysis .

The schema is then harmonized across different datasets, specifically yellow taxi and HVFHV trips, by aligning variable names and definitions. For example, equivalent fields like trip distance and tip amounts are standardized so that both datasets share a common structure. This ensures that meaningful combination and consistent analysis across service types can be performed. Time-based features and policy indicators are also engineered. Timestamps are converted into clean date variables and create a post indicator based on the policy implementation date (January 5 2025). This prepares the data for difference-in-differences analysis by embedding the treatment structure directly into the dataset.

In total, the dataset contains approximately 573 million trips, which would allow for a high-powered analysis. Overall, the dataset is heavily dominated by HVFHV (for-hire vehicle) trips, which account for about 483 million observations, compared to about 90 million yellow taxi trips. As a result, they are separately analyzed by service types in the difference-in-difference as well to account for casual effects by service type.

The overall summary statistics for the data are included in Table 1.

Table 1. Descriptive Statistics of Mobility Dataset

Panel A: Overall Summary

Metric	Value
Total trips	572,952,454
Date Range	2024-01-01 to 2025-12-31
Number of Locations (Taxi Zones)	263

Panel B: Service Composition

Service Type	Trips	Share (%)
Yellow Tax	89,892,322	15.69
HVFHV	483,060,132	84.31

Panel C: Trip Characteristics

Metric	Value
Average Distance (Miles)	5.20
Median Distance (Miles)	2.75
Average Tip (\$)	1.48
Average congestion fee (\$)	0.51

Panel D: Pre vs Post Policy Distribution

Group	Trips	Average Distance (Miles)
Pre(HVFHV)	241,948,390	5.08
Post(HVFHV)	241,111,742	5.03
Pre (Yellow)	41,533,816	4.97
Post(Yellow)	48,358,506	6.86

Panel E: Treated vs Control

Group	Mean Trips	Standard Dev	Observations
Control	7,718	12,346	40,848
Treated	19,686	14,881	13,091

Methodology

Data Pipeline

1. *Data Ingestion (AWS S3)*. Monthly TLC Parquet files were downloaded (yellow, green, FHV, HVFHV) for January 2024 through December 2025 and staged in an S3 bucket partitioned by service type and month. The TLC taxi zone lookup was staged alongside the trip data to support spatial joins. Parquet is retained as the storage format because of its columnar compression and compatibility with Spark and Snowflake.
2. *Data Transformation (Snowflake)*. S3 bucket was connected with raw data via Snowflake. Each trip is joined to the taxi zone lookup to classify pickups and dropoffs as CPZ, Manhattan-above-60th, or outer-borough. An origin-destination trip-type flag is constructed, a post-treatment indicator ($\text{date} \geq 2025-01-05$) is added, and trips are aggregated to the daily level.
3. *Analytical Querying (Snowflake)*. Curated aggregates were loaded into Snowflake for exploratory analysis and pre-trend checks using distributed SQL.
4. *Causal Modeling (Python)*. The aggregated daily series were pulled into Python, where difference-in-differences specification was estimated using statsmodels, test parallel pre-trends, and run robustness checks across service types.
5. *Visualization and Delivery (Python & R)*. Event-study plots, pre-trend diagnostics, and choropleth maps of affected zones were produced in the notebook and embedded in a [GitHub](#) Pages [project website](#) alongside the code and paper.

Experimental methodology

For analysis, Difference in Difference (DiD) quasi-experimental design is used because there is a clear and defined date when treatment occurred. For treatment, each trip that terminated within the CPZ zone was defined as being affected by the increased tax. This approach does have the potential to miss trips that would have crossed the CPZ area without actually terminating there, but the assumption is that this would not be a significant number of trips since there are better ways to complete trips across both the rivers.

The regression specification below takes the logged trip volume for each week from each taxi zone against treatment, post, and dummies for location and the week.

$$\ln(\text{trip volume}) = \text{treatment} + \text{post} + c(\text{location}) + c(\text{week})$$

To ensure validity of the methodology, the parallel lines assumption was verified as shown in Figure 1. Prior to the policy, both the treatment and control groups were displaying a relatively flat trend, making the approach methodologically sound.

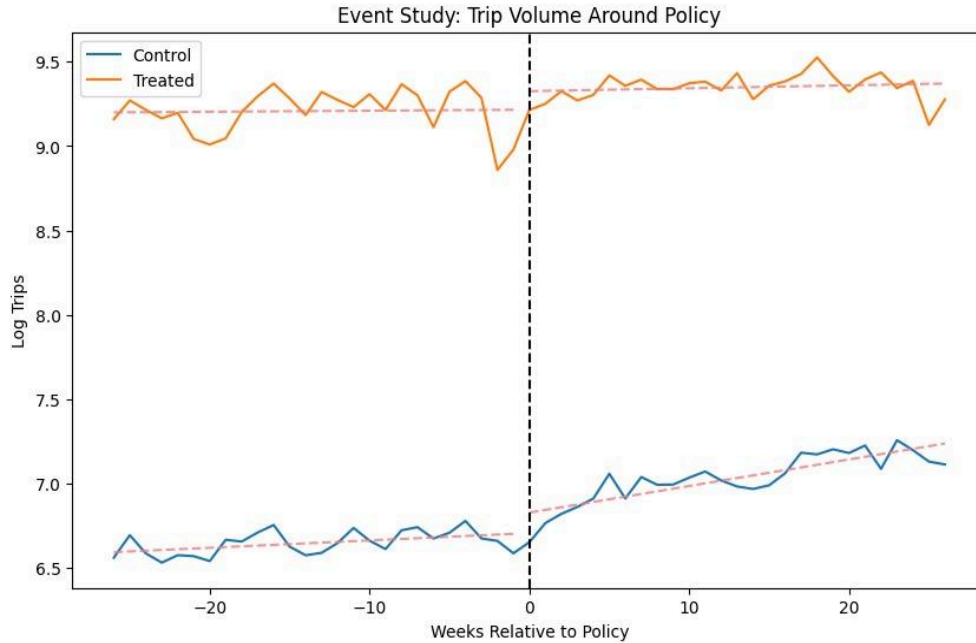


Figure 2. Event Study pre- and post-trend lines for robustness checks

Finally, to look at the effect between Yellow Taxis versus High Volume for Hire Vehicles, the same regression specification was done for each of the mentioned subsamples.

Findings

Regression results

After modeling, statistically significant results were found due to the massive scale of the data, which can be seen in the tables below.

Table 1. Difference-in-Differences Estimates of the Effect of Congestion Pricing on Trip Outcomes for the Full sample of Yellow cabs and HVFHV.

Variable	Coefficient	Standard Error	p-value	% Effect
Treated × Post	-0.3321	0.0421	<0.001	-28.26
Treated	5.3398	0.1822	<0.001	20746.66
Post	0.4947	0.0965	<0.001	64.00

Across ~560 million trips analyzed from January 2024 to December 2025, for-hire vehicle dropoffs inside the Congestion Relief Zone fell by 28.3% after the policy took effect, relative to trips outside the zone.

Table 2. Difference-in-Differences Estimates of the Effect of Congestion Pricing on Trip Outcomes for Yellow Cabs only

Variable	Coefficient	Standard Error	p-value	% Effect
Treated × Post	−0.671	0.008	<0.001	−48.91%
Treated	4.125	0.028	<0.001	6088.64%
Post	1.155	0.019	<0.001	217.56%

Yellow taxis experienced the steepest drop of any vehicle type, nearly half of all CRZ trips lost. The higher base fare combined with the surcharge made yellow cab trips into Manhattan's core significantly less attractive, with many riders likely substituting to transit or app-based alternatives. This fact is backed up by the much less drastic fall in trips from ride share apps being in table 3.

Table 3. Difference-in-Differences Estimates of the Effect of Congestion Pricing on Trip Outcomes for HVFHV

Variable	Coefficient	Standard Error	p-value	% Effect
Treated × Post	-0.0932	0.0036	<0.001	-8.90
Treated	8.947	0.0256	<0.001	76862.876
Post	0.165	0.00758	<0.001	17.99

From the 8.9% decline, it can be ascertained that Uber and Lyft trips into the CRZ saw a modest but statistically significant decline, suggesting riders using app-based services are less price-sensitive given the relatively low \$1.50 per-trip surcharge.

Geospatial dispersion of changes

Spatial distribution for declines in taxis traffic from the DiD analysis was also mapped. Figure 2 shows that the most affected parts of New York were concentrated primarily in Manhattan (even areas outside the effect zone) and large swathes of Brooklyn close to the island; it would also be expected to see a similar result for Jersey City across the river.

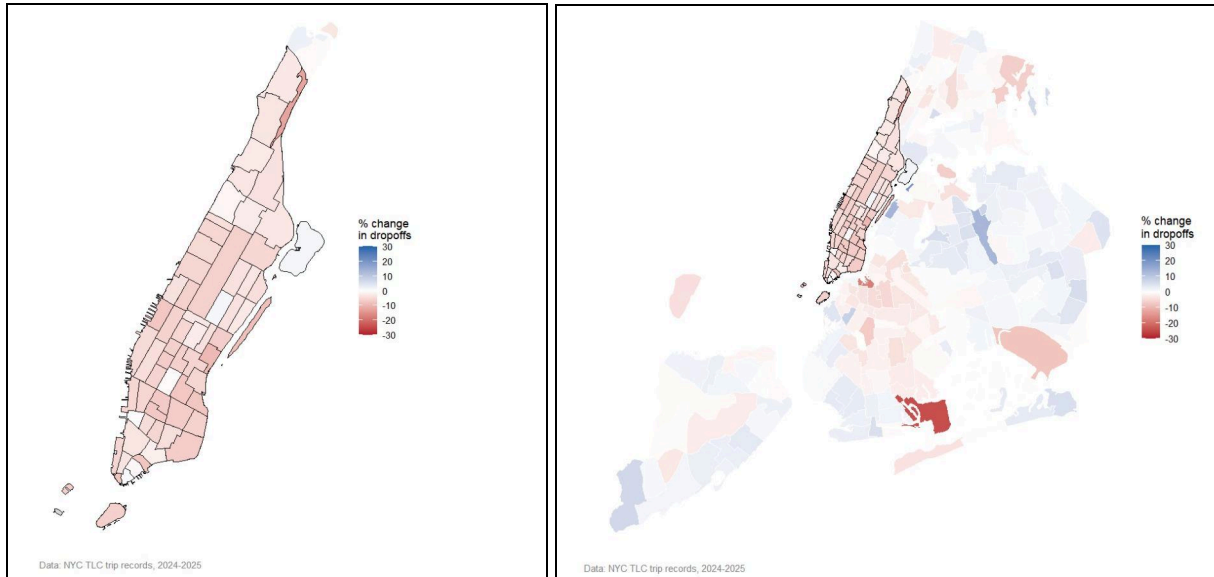


Figure 3. Geospatial dispersion of ridership loss across NYC

Conclusion and further research

This study provides causal evidence that New York City's congestion pricing policy meaningfully reduced for-hire vehicle trips into the Congestion Relief Zone. Analyzing approximately 560 million trips across 2024 and 2025, the difference-in-differences estimates show a 28.3% decline in dropoffs inside the CPZ relative to zones outside it. This effect is not uniform across vehicle types. Yellow taxis experienced a near-50% reduction in CRZ trips, while High Volume For-Hire Vehicles saw a more modest 8.9% decline. The contrast reveals a clear price sensitivity gradient: yellow cab riders, who face higher base fares compounded by the surcharge, appear far more responsive to the added cost than app-based riders absorbing a comparatively small \$1.50 charge. The spatial distribution of these effects concentrated in lower Manhattan and adjacent Brooklyn neighborhoods further suggests that the policy's burden is not evenly distributed across the city's geography.

Taken together, these findings suggest that congestion pricing is achieving its primary goal of reducing vehicle volume in Manhattan's core. However, the heterogeneity across vehicle types raises important questions about who bears the cost of that reduction and where displaced trips are going, questions the data alone cannot answer.

Several extensions would sharpen the policy implications of this work. The most immediate priority is understanding mode substitution. The analysis documents a decline in taxi and FHV trips but cannot distinguish whether displaced riders shifted to subway, bus, biking, or simply made fewer trips. Pairing TLC data with MTA ridership records and Citi Bike usage over the same window, using the same DiD framework, would directly test whether congestion pricing is driving riders toward greener alternatives or suppressing mobility altogether.

Closely related are equity concerns. Yellow taxis disproportionately serve outer-borough and lower-income riders who may lack flexible alternatives to car-based travel into Manhattan. The near-50% drop in yellow cab CRZ trips is a meaningful signal that the policy may be pricing out riders who can least afford substitution costs. Future work should examine trip-level fare data alongside neighborhood income and transit access measures to assess whether the policy's benefits are accruing equitably.

On the environmental side, linking the observed drop in vehicle volume to air quality monitoring data, specifically NO₂ and PM_{2.5} readings from EPA stations near the CPZ boundary, would allow researchers to quantify whether the traffic reduction is translating into measurable health improvements for Manhattan residents. Given the well-documented co-location of pollution burdens and communities of color, this is a particularly valid empirical question.

Finally, the sharp geographic boundary of the CPZ presents an opportunity for a regression discontinuity design, exploiting variation in trip volumes just inside and outside the zone boundary to validate and complement the DiD estimates in this paper. Taken together, these extensions would move the analysis from documenting that congestion pricing reduced for-hire vehicle trips to understanding for whom, at what cost, and with what broader urban consequences.

References

- Croci, E. (2016). Urban road pricing: A comparative study on the experiences of London, Stockholm and Milan. *Transportation Research Procedia*, 14, 253-262. <https://doi.org/10.1016/j.trpro.2016.05.062>
- Kassirer, J., & Boddy, S. (2014). *Stockholm Congestion Pricing*. Tools of Change. Retrieved April 30, 2026, from <https://toolsofchange.com/en/case-studies/detail/670/>
- Metropolitan Transportation Authority. (n.d.). *Congestion Relief Zone*. Retrieved April 6, 2026, from <https://congestionreliefzone.mta.info/>
- Metropolitan Transportation Authority. (n.d.). *Taxis and for-hire vehicles (FHV) tolls*. Retrieved April 6, 2026, from <https://www.mta.info/fares-tolls/tolls/congestion-relief-zone/taxi-fhv-tolls>
- New York City Taxi and Limousine Commission. (n.d.). *TLC trip record data*. Retrieved April 6, 2026, from <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>
- Sager, M. (2025, January 9). *You Wasted \$771 Sitting in Traffic Last Year*. Newsweek. <https://www.newsweek.com/traffic-study-wasted-money-economy-congestion-2012671>
- Singapore-MIT Alliance for Research and Technology. (2021, April 23). *Study finds ride-sharing intensifies urban road congestion*. MIT News. Retrieved April 30, 2026, from <https://news.mit.edu/2021/ride-sharing-intensifies-urban-road-congestion-0423>
- United States Environmental Protection Agency. (2026, April 6). *Carbon Pollution from Transportation | US EPA*. EPA. Retrieved April 30, 2026, from <https://www.epa.gov/transportation-air-pollution-and-climate-change/carbon-pollution-transportation>

Appendix

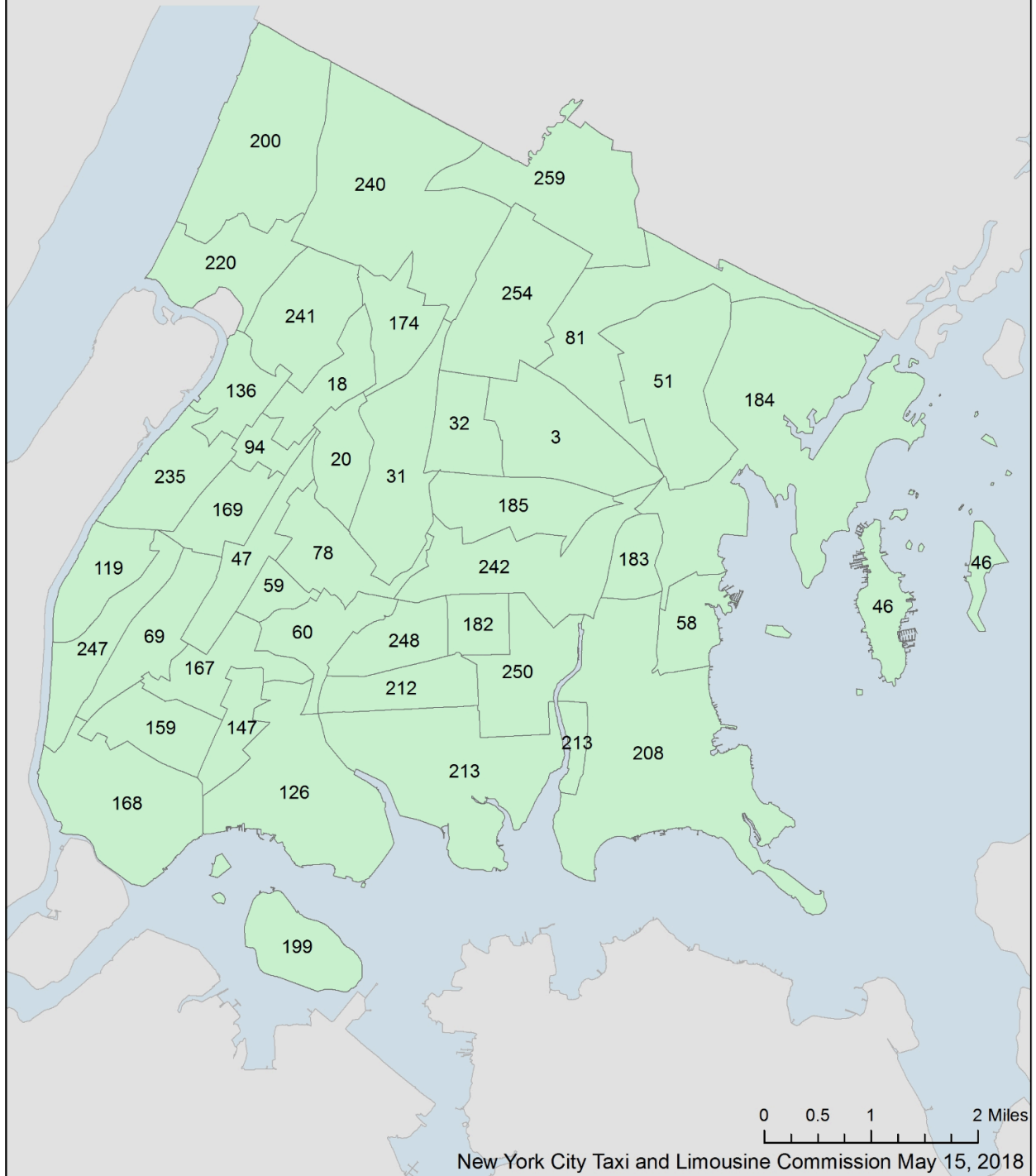
1. Map information

The pick up and drop off locations use a coded map of NYC to track where they were picked up, unfortunately we don't have the exact coordinates. We can see the maps below and the lookup table can be seen here:

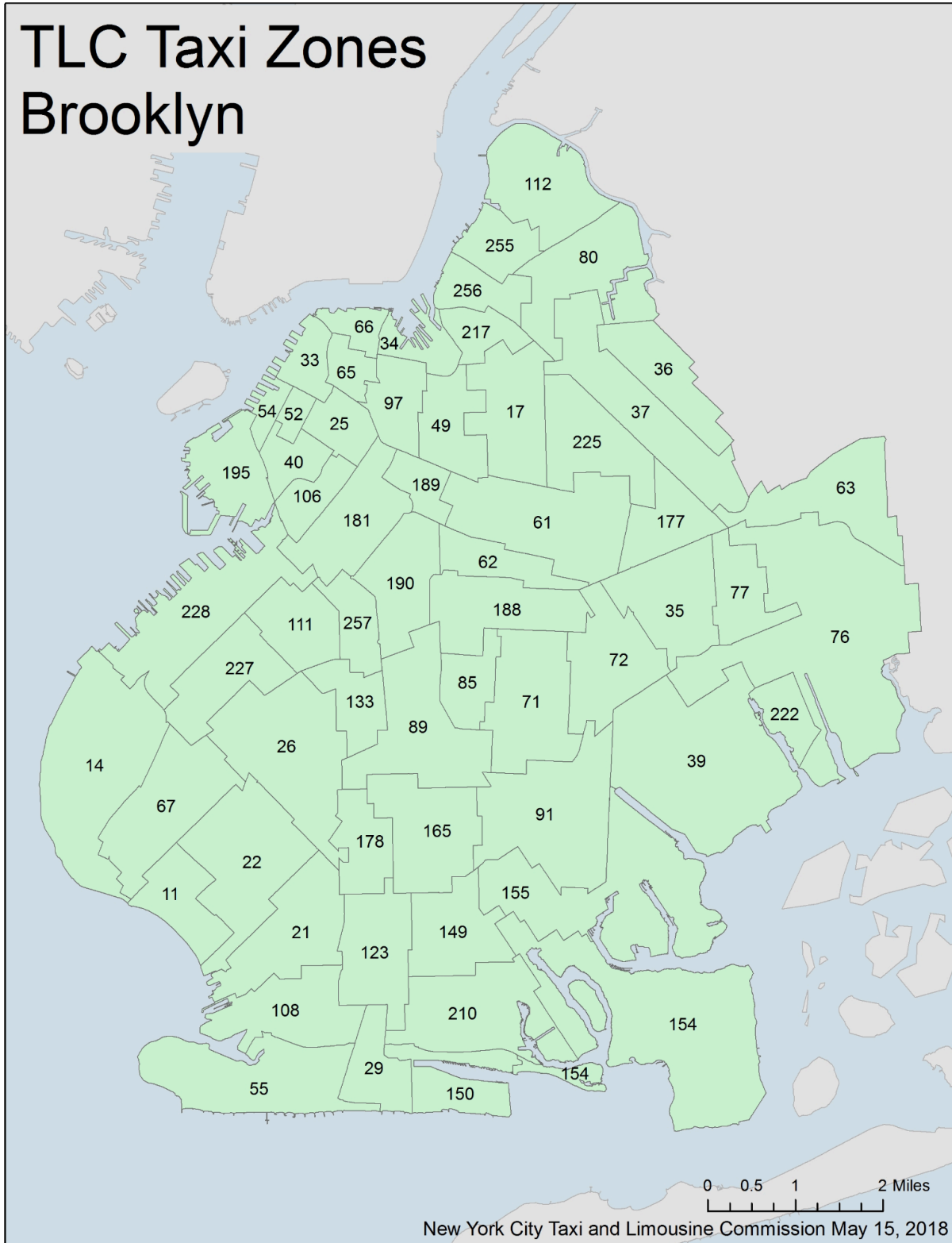
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TLC Taxi Zones

Bronx

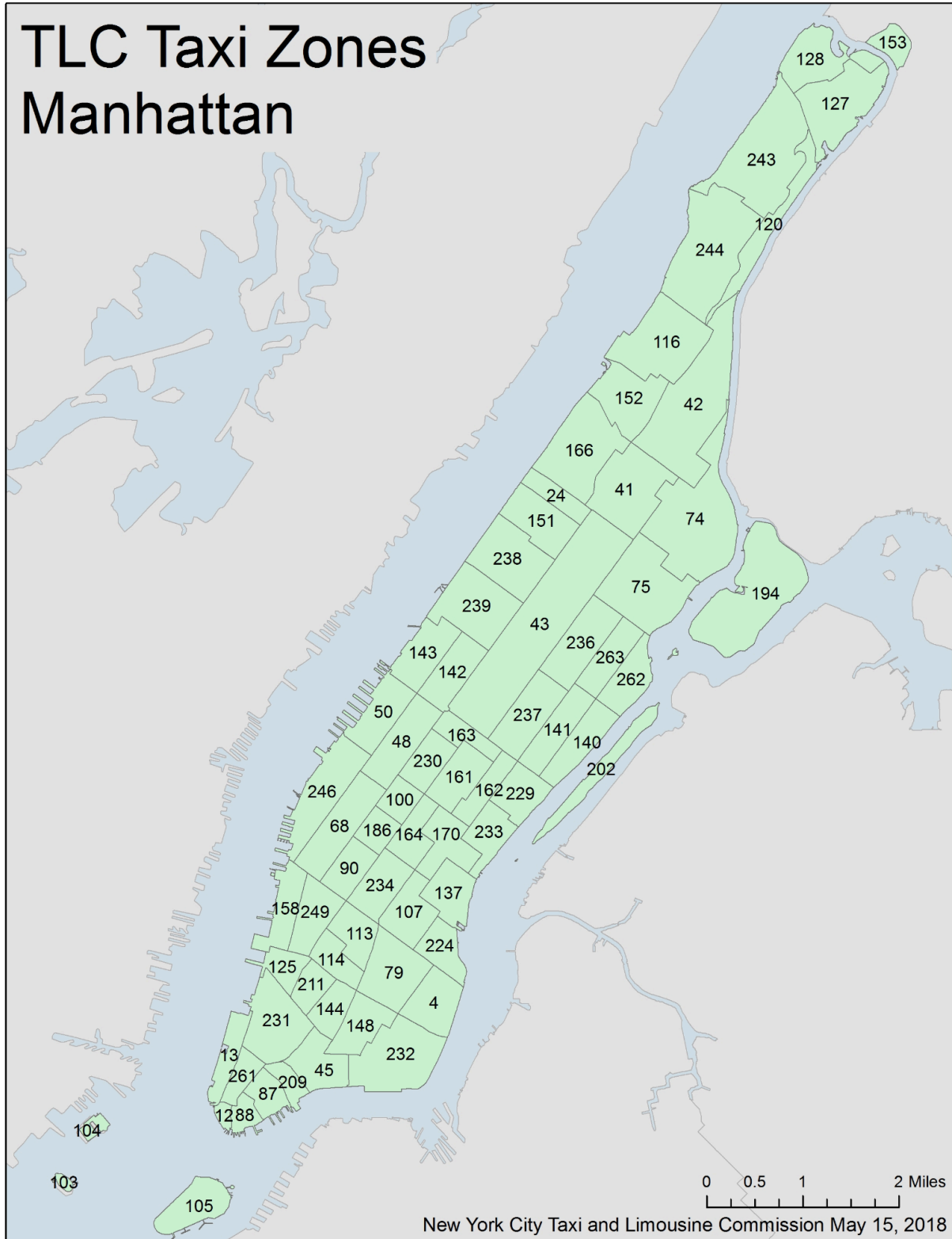


TLC Taxi Zones Brooklyn



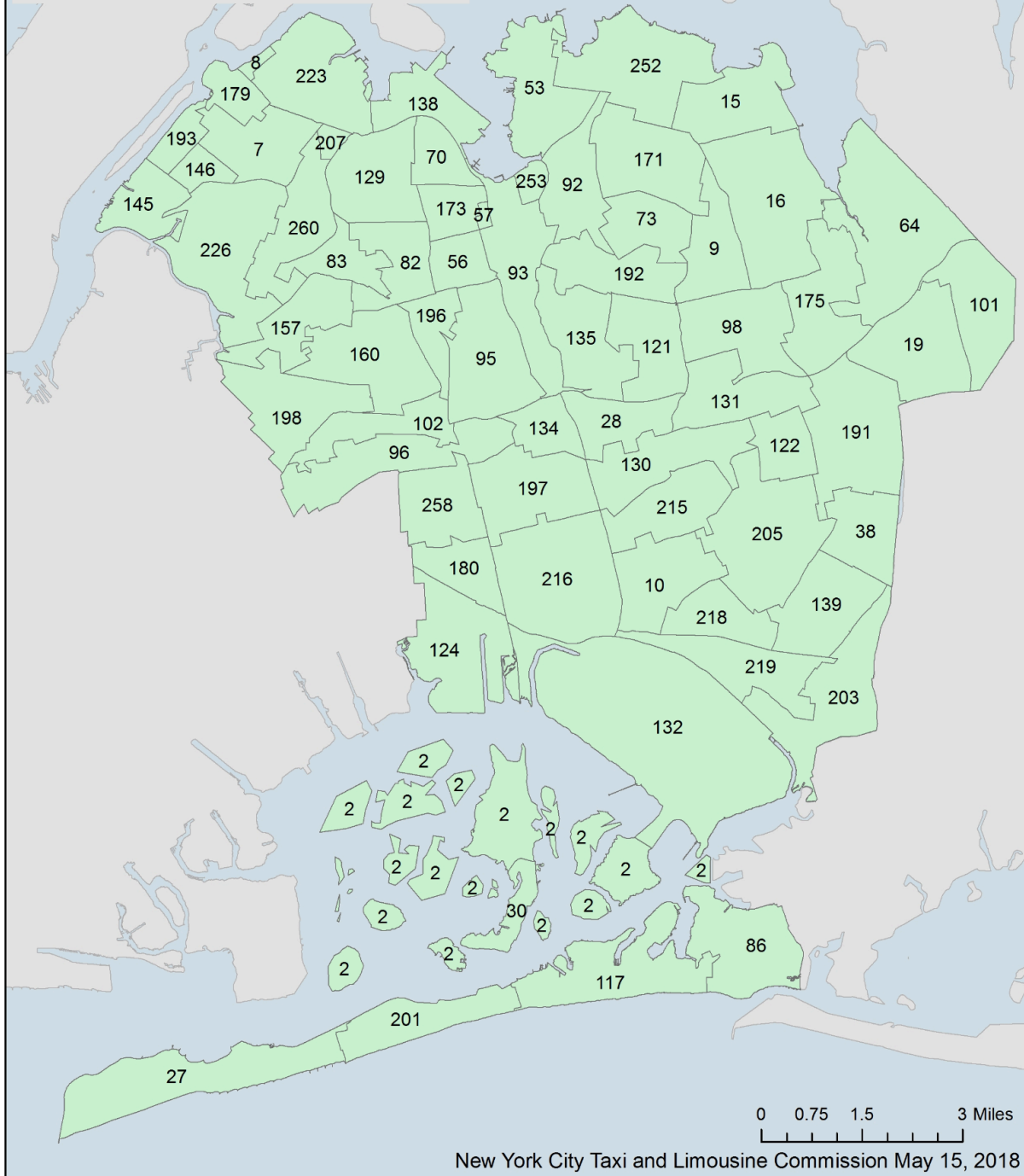
New York City Taxi and Limousine Commission May 15, 2018

TLC Taxi Zones Manhattan



TLC Taxi Zones

Queens



TLC Taxi Zones Staten Island

